

# ALI HAGGAG

SOFTWARE ENGINEER | FINTECH PLATFORMS | ENTERPRISE AI & SYSTEM ARCHITECTURE

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**PROFILE** Software engineer working at the intersection of mission-critical fintech, scalable platform engineering, and enterprise AI. Contributed to InstaPay, Egypt's leading instant payment platform, with hands-on experience in Spring Boot microservices, PostgreSQL, Redis, React, and Python. Proficient in performance engineering, system design, and project management, with a strong ability to translate business requirements into secure, maintainable systems.

## CAREER IMPACT

### NATIONAL-SCALE FINTECH

Contributed to the engineering of InstaPay, a critical part of Egypt's digital payments ecosystem, supporting millions of active users and high-volume financial activity.

### MISSION-CRITICAL ENGINEERING

Designed and implemented services where availability, correctness, security, and predictable performance directly affect customers' financial operations.

### PLATFORM EVOLUTION

Helps shape backend services, data access patterns, caching, and service architectures so the platform can sustain growth without sacrificing maintainability.

### BUSINESS-TO-TECHNOLOGY TRANSLATION

Bridges operational constraints and business objectives into practical architecture, implementation plans, and project coordination.

### CROSS-STACK OWNERSHIP

Can reason across backend services, databases, APIs, frontend experiences, and delivery concerns instead of optimizing one layer in isolation.

### AI TRANSFORMATION MINDSET

Applies enterprise architecture, governance, developer experience, and project thinking to emerging AI systems rather than treating AI as a standalone feature.

## CORE EXPERTISE

| Enterprise Software Architecture    | Distributed & High-Availability Systems | Fintech & Payment Platforms              | AI Systems & LLM Integration |
|-------------------------------------|-----------------------------------------|------------------------------------------|------------------------------|
| Spring Boot Microservices           | API & Service Design                    | PostgreSQL & Performance Engineering     | Redis & Caching Strategies   |
| React / Next.js Product Engineering | Platform Reliability & Scalability      | Technical Discovery & Trade-off Analysis | Product-Minded Engineering   |

## PROFESSIONAL EXPERIENCE

### Egyptian Banks Company - Cairo

Nov 2023 - Present

#### Software Engineer

- Contributed to InstaPay, Egypt's leading instant payment application, helping sustain a trusted national payments experience for millions of customers.
- Designed and deployed Spring Boot microservices for high-volume fintech workloads, balancing performance, modularity, deployment safety, and long-term maintainability.
- Strengthened platform scalability by improving service behavior, PostgreSQL access patterns, indexing, caching, and load distribution as transaction volume grows.
- Investigated performance and reliability issues across application and data layers, turning problem areas into reliable engineering improvements rather than one-off fixes.
- Worked within strict financial services expectations for security, correctness, availability, and operational resilience.
- Collaborated across engineering and business teams to convert complex requirements into clear technical designs and project-ready implementations.
- Contributed to architectural decisions affecting service architectures, integration patterns, platform reliability, and future growth.

Frontend Developer

- ;eli,ere) 5eact an) 6e1t7s pro) /cts that translate) client an) . /siness re=/irements into fast, responsi,e, maintaina.le )igital e1periences'
- ;esigne) re/sa.le fronten) fo/n)ations an) component patterns that impro,e) consistency an) allowe) teams to )eli,er new pages an) feat/res more e>ciently'
- <or(e) closely with .ac(en) engineers to shape API integrations, )ata ?ows, error han)ling, an) /ser interactions as one coherent pro) /ct e1perience'
- Impro,e) percei,e) an) meas/re) performance thro/gh ren)ering, loa)ing, responsi,eness, an) client+si)e optimi:ation'

ENTERPRISE AI & APPLIED AI EXPERTISE

Applie) architect/re an) 5& ; e1perience across enterprise AI platforms, local inference, )oc/ment intelligence, co)ing assistants, an) agentic wor(?ows'

- **Enterprise AI platform architecture:** ;esigne) a m/lti+ser,ice platform mo)el co,ering i)entity, wor(spaces, AI ser,ices, )ata ser,ices, API gateway, mo)el access,go,ernance .o/n)aries, an) role+speci-c AI e1periences'
- **AI-powered engineering workflows:** ;esigne) m/lti+agent wor(?ows for )e,elopment,=/ality engineering, ;e,@ps, information sec/rity, )oc/mentation, test co,erage, an) co)e+change s/pport'
- **RAG & document intelligence:** <or(e) thro/gh retrie,al, ch/n(ing,em.e))ings, )oc/ment in)e1ing, conte1t assem.ly, so/rce gro/n)ing, an) pri,ate enterprise (nowle)ge+assistant patterns'
- **Local and on-device inference:** E,al/ate) local 44A /se on Apple Silicon /sing @llama an) A4B, incl/)ing =/anti:e) mo)els, pri,acy+-rst wor(?ows, har)ware constraints, an) mo)el+=/ality tra)e+o8s'
- **Coding-assistant architecture:** E1plore) I ; E+integrate) AI assistants, inference ser,ers, pre-1\*CD caching, latency targets, tool integration, an) mo)el e,al/ation for practical )e,eloper /se'
- **Agentic systems & tool orchestration:** ;esigne) wor(?ows in ,ol,ing speciali:e) agents, str/ct/re) o/tp/ts, tool e1ec/tion, appro,al gates, temporary .ranches, p/ll re=/ests, an) h/man re,iew'
- **AI governance & security:** Applies enterprise concerns s/ch as a/thentication, a/thori:ation, tenant isolation, a/ )ita.ility, )ata pri,acy, sec/re mo)el access, an) controlle) e1ec/tion'

TECHNICAL EXPERTISE

|                         |                                                                                                     |
|-------------------------|-----------------------------------------------------------------------------------------------------|
| Backend & APIs          | Ea,a, Spring 2oot, E1press7s, 5ES9 APIs, Fraph34, microser,ices                                     |
| Frontend                | 5eact7s, 6e1t7s, 9ypeScript, Ea,aScript, 5e) /1, 9ailwin) CSS                                       |
| Data & Performance      | PostgreS34, S34, in)e1ing,=/ery optimi:ation, 5e)is, caching, performance t/ning                    |
| AI & Inference          | 44A integration, 5AF patterns,em.e))ings, agent wor(?ows, @llama, A4B, local+mo)el e,al/ation       |
| Architecture & Delivery | System )esign, )istri./te) systems, scala.ility, relia.ility, sec/re co)ing, ;oc(er, Agile )eli,ery |

EDUCATION & PROFESSIONAL DEVELOPMENT

|                                                                     |                                                                                                                                                                                                |
|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| German University in Cairo<br>2'Sc' Architect/ral Engineering, 2 2! | Professional Development<br>Java Design Patterns & SOLID Principles<br>Principles of Secure Coding<br>Agile Project Management<br>AWS Machine Learning Engineer - Associate path (in progress) |
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